

Miriam Lerma

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Research interest

I am a ecology professional with strong foundation in monitoring and conservation, particularly with seabird populations. My experience spans academic research and applied conservation work with NGOs, where I have developed the ability to collaborate effectively across diverse international teams and organizational settings. I have a strong analytical background, with a publication record that reflects my ability to assess and synthesize complex information effectively. One of my key strengths is my proficiency in R programming, which I use to develop custom tools that streamline analysis workflows and enhance efficiency.

Relevant skills

Fieldwork	Laboratory	Computer	Languages
Catching and handling wild animals – Sample collection: extraction blood & feathers – Censusing populations – monitoring breeding success	Heavy metal analyses – Stable isotopes – Plasma metabolites analyses	R - SQL – QGIS Statistics – SPSS Sigmaplot Git – Github Word	Spanish (mother tongue) – English (toefl ibt 99) – German (b1)

Education

Doctoral studies Christian-Albrecht-Universität zu Kiel	2015-2020 Kiel, Germany
Master in science in marine biology Universidad Nacional Autonoma de México	2011-2013 Mazatlán, Mexico
Bachelor in science in biology Universidad de Guadalajara	2005-2009 Guadalajara, Mexico

Work experience

Research associate Dachverband Deutscher Avifaunisten • Project: OWP Seevögel II	2026-current
Research associate Dachverband Deutscher Avifaunisten • Project: Marbird	2021-2026
Postdoctoral researcher Forschung und Technologie-Zentrum Westküste (FTZ, Kiel University) • Project: TopMarine	2021
Field technician Fondo Noroeste AC (civil association) • Project: Composition, abundance and habitat use of birds on shrimp farms on Bahia de Santa Maria, Sinaloa	2015
Field technician CONANP (National Commission of Natural Protected Areas) • Project: Biological monitoring on natural protected areas (focused on shorebirds)	2014
Field technician CONANP (national commission of natural protected areas) • Project: Conservation of species on risk (focused on American oystercatcher)	2014
Advisor UNAM-FONATUR (National autonomous university of Mexico- National trust fund for tourism development) • Subproject: Fauna monitoring on project CIP pacific coast on Escuinapa, Sinaloa	2014
Instructor	2014

CONANP-IGC (National commission of natural protected areas-Gulf of California islands)	
• Course: Introduction to studies on birds	
Field technician	2012
Sonoran joint venture (Binational bird conservation)	
• Project: Monitoring seabirds in Santa Maria bay, Sinaloa: Population size, nesting habitat, reproductive success and diet	
Field technician	2012
CONACyT (National council on science and technology)	
• Project: Seabirds as environmental indicators in coastal systems of economic importance	
Field technician	2012
Grupo de ecología y conservación de islas A.C.	
• Monitoring eradication of invasive mammals: Farallón de San Ignacio and San Pedro Martir islands	
Field technician	2010
Fondo Mexicano de Conservación de la Naturaleza	
• Project: ecological and ecotoxicological monitoring of the seabird community of Santa maria bay: establishing base lines for the use of bioindicators in decisions of environmental management, Sinaloa	
Technical assistant	2009
Resources from basic science research SEP-CONACyT and PAPIIT	
• Project: Morphologic, isotopic and genetic characterization of <i>Calidris mauri</i> in the northwest of México	
Education intern	2009
Pronatura river of raptors, Cardel, Veracruz	
• Orientation to school groups, local people, and visitors	
Technical assistant	2009
Ducks unlimited de México (Binational bird conservation)	
•Project: evaluation of the waterbird community in rehabilitated mangroves sites	
Collaborator	2008
University of Guadalajara	
•Course: Introduction to the study of birds	
Peer-review publications	
First report on habitat use and trip parameters of Yellow-footed Gulls	2026
<i>Larus livens</i> during the breeding and post-breeding period	J Ornithol
Lerma M, Soldatini C, Alcalá-Santoyo JE, López-Hernández JA, Albores-Barajas YV, Garthe S	
Urban and Wild Kelp Gulls: Tracking Seasonal Shifts in Habitat Use and Movement Patterns	2026
Lerma M, Seguel M, Fernandez CE, Garthe S, Luna-Jorquera G	Eco Evo 16: e72974
Variations in inter-specific and sex-related niche partitioning in pelagic boobies during their annual cycle	2024
Lerma M, Dehnhard N, Castillo-Guerrero JA, Hernandez-Vazquez S, Voigt CC, Garthe S	Eco Evo 14: e11255
Corticosterone levels, leukocyte profiles, and foraging and diving behaviours of Humboldt penguins during chick rearing in northern Chile	2023
Lerma M, Villavicencio C, Luna N, Portflitt-Toro M, Serratos J, Luna-Jorquera G, Garthe S, Quispe R	Emu 123: 71-78
Seabird morphology determines operational wind speeds, tolerable maxima, and responses to extremes	2023
Nourani E, Safi K, De Grissac S, Anderson DJ, Cole NC, Fell A, Gremillet D, Lempidakis E, Lerma M, McKee JL, Pichegru L, Provost P, Rattenburg NC, Ryan PG, Santos CD, Schoombie S, Tatayah V, Weimerskirch H, Wikelski M, Shepard ELC	Current biology 33: 1179-1184
Nutritional state variations in a tropical seabird throughout its breeding season	2022
Lerma M, Dehnhard N, Castillo-Guerrero A, Fernandez G	J comp phy b 192: 775-787
Breeding stage, not sex, affects foraging characteristics in masked	2020

boobies at Rapa Nui

Lerma M, Dehnhard N, Luna-Jorquera G, Voigt CC, Garthe S

Beh ecol soc 74: 149

Foraging ranges of Humboldt penguins *Spheniscus humboldti* from Tilgo island: the critical need for protecting a unique marine habitat.

2020

Mar ornithol 48: 205

Quispe R, Lerma M, Luna N, Portflitt-Toro M, Serratos J, Luna-Jorquera G

Foraging ecology of a marine top predator in the eastern tropical pacific over 3 years with different ENSO phases

2020

Mar biol 167: 88

Lerma M, Castillo-Guerrero JA, Hernandez-Vazquez S, Garthe S

Zinc concentrations in Blue-footed booby (*Sula nebouxii*) eggs, nestlings, and adults

2020

J sea research 165: 101952

Lerma M, Castillo-Guerrero JA, Garcia-Hernandez J, Fernandez G

Foraging ecology of masked boobies (*Sula dactylatra*) in the world's largest 'oceanic desert'

2020

Mar biol 167: 87

Lerma M, Serratos J, Luna-Jorquera G, Garthe S

Non-breeding distribution, abundance, and roosting habitat use of the American oystercatcher (*Haematopus palliatus frazari*) in Sinaloa, Mexico

2017

Waterbirds 40: 95-103

Lerma M, Castillo-Guerrero JA, Palacios E

Lead, cadmium and mercury in the blood of the Blue-footed booby (*Sula nebouxii*) from the coast of Sinaloa, Gulf of California, Mexico

2017

Mar poll bull 110: 293-298

Lerma M, Castillo-Guerrero JA, Ruelas-Inzunza J, Fernandez G

Environmentally-mediated flexible foraging strategies in Brown boobies in the Gulf of California

2016

Ardea 104: 33-57

Castillo-Guerrero JA, Lerma M, Mellink E, Suazo-Guillén E, Peñaloza-Padilla EA

Conferences

2025	Deutschen ornithologen-gesellschaft	Erfurt, DE
2023	Pacific seabird group 50th annual meeting	San Diego, US
2022	Deutschen ornithologen-gesellschaft	Wilhelmshaven, DE
2021	7th annual bio-logging science symposium	Hawaii, virtual
2018	Pacific seabird group 45th meeting	La Paz, MX
2017	VI reunión internacional de investigadores del Archipiélago Revillagigedo	Colima, MX
2017	41th waterbirds 2017	Reykjavik, IS
2015	Waterbirds 2015: challenges and responses	Bar harbor, US
2013	Pacific seabird group 40th annual meeting.	Portland, US
2011	Congress for the study and conservation of the birds in Mexico	Mazatlán, MX

Trainings

2024	Via mento International University of Kiel. Kiel, Germany
2023	Working with the IACUC (Institutional Animal Care and Use Committee) Oregon University. Online, US
2022	Risk and impact assessment course EUROWA. Texel, Netherlands
2021	From zero to (s)hero: five-part introduction to R R-ladies Freiburg. Online, Germany
2020	Programación en R para estudios de biologgging de vertebrados marinos y sus interacciones con la actividad pesquera en el Perú Universidad Científica del sur. Online, Peru
2018	ANIMOVE Max Planck institute. Radolfzell, Germany
2018	Basics in university teaching

- Integrated school of ocean sciences. Kiel, Germany
2016 International summer school on stable isotopes in animal ecology
Centre for stable isotope ecology. Berlin-Brandenburg, Germany
- 2016 GIS for marine sciences**
Integrated school of ocean sciences. Kiel, Germany
- 2016 Analysis of univariate data sets - first steps in R**
Integrated school of ocean sciences. Kiel, Germany